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Causes of the Underrepresentation of Women in STEM

Despite women pursuing STEM at higher rates than in previous decades, STEM fields remain disproportionately male (Tian et al., 2022). At UVic, I have noticed that only around 20% of students in my Computer Science courses are women and that there are culture issues within the department. Every woman I know who is studying Engineering or Computer Science has had some experience of being told that she does not belong there, not being given technical work in group projects, or being the only woman in her class. This paper seeks to investigate the causes of this underrepresentation of women in STEM by exposing the variety of factors at every stage of a woman's life that may make STEM unappealing or inaccessible to her. I start by evaluating the claim that higher representation of men in STEM is due to biological differences in math ability. Then, looking at factors that impact girls in elementary or secondary school, I will discuss the impact of classroom environment on girls' grades, and how girls' math performance is related to their female teachers' math anxiety. Finally, I will discuss the post-secondary STEM environment and what awaits women who do decide to pursue STEM.

Biological Factors

Benbow and Stanley are researchers who founded "The Study of Mathematically Precocious Youth" (*Study of Mathematically Precocious Youth* [SMPY], n.d.). Over 5000 intellectually gifted participants were identified between 1972 and 1997, and the group is continuing to complete a 50-year longitudinal study on these participants (SMPY, n.d.). During these searches for mathematically precocious youth, the researchers identified more boys than girls, which led to the group searching for reasons for this gender difference (Benbow & Stanley, 1980). In an influential and controversial paper, Benbow and Stanley (1980) claim that sex differences in achievement in math are a result of "superior male mathematical ability" (p. 1264)

which could be related to male advantage in spatial skills, and that gender socialization processes are unlikely to be the only explanation for this observed sex difference.

This paper was highly publicized and resulted in mother's who had seen the publicity rating their daughter's math abilities lower, which has been shown to influence the daughter's own thinking about her math abilities (Hyde, 1997). A meta-analysis conducted by Hyde (1997) showed no performance differences in math between general population boys and girls, which casts doubt on Benbow and Stanley's conclusion that performance differences between mathematically precocious boys and girls are due to biological factors. In Benbow and Stanley's (1980) paper they say that 43% of the participants in their study were girls and that there was no significant difference in math grades between the boys and girls in school. However, when the students took the SAT during recruitment, on average four to five years younger than when students typically take the SAT, boys significantly outperformed girls (Benbow & Stanley, 1980). Performance differences on the SAT have been linked to strength of spatial skills in other studies (Tian et al., 2022).

Spatial skills are a strong predictor of earning a STEM degree in the future, more so than math ability (Tian et al., 2022). Spatial skills are "abilities involved in mentally manipulating objects in space and reasoning about spatial relationships" (Tian et al., 2022, p. 2). It has been observed that males outperform females on many spatial tasks, with these differences being noticed as early as age six (Tian et al., 2022). When strong spatial skills predict earning a STEM degree in the future, and boys outperform girls on spatial skills, it tracks that girls would be underrepresented in STEM. Benbow and Stanley (1980) claim that this difference in spatial skills is biological, but recent work by Kung and Hines (2025) suggests that strength of spatial skills is related to the type of play that children engage in. Parents were asked to report the type of play

their child engaged in at age 3.5 years by rating how often their child engaged in a variety of different play activities that are typically gendered. The children then did a spatial skills test at age 13. It was found that both boys and girls who engaged in masculine play as children outperformed boys and girls who engaged in feminine play. However, within the masculine play and feminine play groups, boys still outperformed girls. This suggests that spatial skills are influenced by the type of play that children engage in, which is typically gendered. Given that the spatial skills test was administered almost ten years after parents reported the type of play their child engaged in, it is reasonable to assume that other factors in the intervening years may have contributed to the performance difference still observed between boys and girls.

Early School Factors

The school environment can strongly impact the interest and performance of girls in STEM classes. The size of the gender gap in math varies around the world and is strongly correlated with women's position in society (Salikutluk & Heyne, 2017). So, the strength of belief in traditional gender roles and masculinity norms in classrooms could impact the gender gap in math observed within that classroom. Salikutluk and Heyne (2017) found that the strength of class belief in traditional gender roles, measured by belief in men being the breadwinner and women being responsible for childcare and the household, did not impact the grades of girls in the class. Stronger class belief in traditional masculinity norms, measured by level of agreement with statements about a man using violence to protect his family, showed decreased performance in girls. In classes with low belief in traditional masculinity norms, gender gaps in math performance were small or even favoured girls. Interestingly, as the strength of class belief in traditional masculinity norms increases, boys' performance increases and girls' performance

decreases. So not only are girls' grades negatively impacted by stronger class belief in traditional masculinity norms, but boys' grades also actually benefit from this environment.

Math anxiety is a negative reaction experienced when subjected to mathematical problem solving (Maloney et al., 2013). It has been shown that the level of math anxiety in a female teacher predicts the math performance of the girls in her class, but not the boys (Inzlicht & Schmader, 2012). Specifically, the more math anxiety a female teacher had, the lower the student's math achievement was (Maloney et al., 2013). At the end of the year with a math-anxious teacher, her female students were more likely to agree with the stereotype that girls are good at reading and boys are good at math (Maloney et al., 2013). Though these studies do not claim that girls directly inherit math anxiety from their female teachers, it is considered to be a risk factor for girls developing math anxiety (Maloney et al., 2013). It has also been shown that those with stronger spatial skills report feeling less math anxiety (Tian et al., 2022). Given that people with math anxiety tend to avoid math intensive careers (Maloney et al., 2013), which would include many STEM fields, early susceptibility of developing of math anxiety could certainly be pushing girls away from STEM.

Post-secondary STEM Environment

Stereotype threat is a phenomenon where people belonging to a certain group will perform worse than they are capable of when they are aware of a negative stereotype related to their group (Inzlicht & Schmader, 2012). People are impacted by stereotype threat to differing degrees, with people who identify more strongly with a certain group and those who care more about success being more strongly impacted (Maloney et al., 2013). Stereotype threat does not only impact women's test scores but also impacts their feelings of belonging in STEM. Members of marginalized groups will often search for situational cues that their identity is not a liability in

their current environment (Inzlicht & Schmader, 2012). If, when searching for these cues, the individual finds identity-safe cues then that signals that their identity is valued and welcome in that environment, which could reduce the stereotype threat effect (Inzlicht & Schmader, 2012). Alternatively, finding identity-threatening cues will initiate a vigilance process where marginalized individuals will continue to look for other clues that their identity might be a liability (Inzlicht & Schmader, 2012). In my experience, women I have met in the Engineering and Computer Science faculty very strongly identify with the fact that they are a woman in a male dominated field and care deeply about proving that they are just as good as the male students. This means these women are particularly prone to stereotype threat effect. Over time, this identity threat can impact women's performance and desire to continue in STEM (Boucher et al., 2023). One suggestion to ensure that women can find identity-safe cues is to highlight diverse scientists in the classroom, such as female scientists or scholars (Boucher et al., 2023).

A study by Becker & Nilsson (2021) looked at ten textbooks used in undergraduate chemistry courses published between 2016 and 2020. They found that across the ten textbooks there were a total of 360 unique names of STEMM (Science, Technology, Engineering, Mathematics, and Medicine) professionals; 12 of them were women. In fact, one textbook had no female STEMM professionals listed in the index, and another misspelled the name of a female scientist. Despite having a total of 360 unique names, only 8 of these scientists, all of whom are male, were named in all ten textbooks. This implies that the other selection of predominantly male scientists in each textbook may not be essential and could be replaced with female scientists to ensure that female students will have representation (Becker & Nilsson, 2021). Not only were women underrepresented in the STEMM professionals listed in the textbook, but they also made up only 30% of human images in the textbook. These images were not only of

scientists, but included people playing sports, at work, and receiving medical care. This representation has also not improved over time. Four of the ten textbooks had first editions published between 1972 and 1987 and substantial changes have been made between the first and current editions. However, the number of women decreased in two books, stayed the same in one, and increased in one. So, not only are female students not seeing representation of female scientists, they are likely also noticing the overall lack of women in their textbooks which could be signalling to them that they do not belong.

Simply having women present in textbooks is not enough, as the quality of the representation is also important. In a textbook being used in one of my courses this term, Robert Vanderbei's *Linear Programming: Foundations and Extensions* (5th edition, 2020), there are some upsetting depictions of women. In a proof of König's Theorem using flow networks, we "Assign one unit of supply to each girl node and a unit of demand to each boy node" (Vanderbei, 2020, p. 248). Not only is the boys and girls marrying each other language outdated (though still often used), but this setup implies that girls have something that boys want, which is a common misogynistic idea at the centre of incel ideology. Another example, this time a practice exercise in the same textbook, describes two men and two women who are stranded on a desert island and starving. The two men would like to "give their affection" (Vanderbei, 2020, p. 268) to one of the two women, giving them food as a trade. He goes on to say that the two women, "being proper young women" (Vanderbei, 2020, p. 268) will only accept affection from one of the two men. This question is clearly deeply inappropriate, and it is concerning that it exists in a textbook published as recently as 2020 and being used in a course in 2026. Any women in my class who read this question will absolutely be receiving an identity-threatening cue that they are not respected here. Fortunately, the class environment has included many identity-safe cues, and the

instructor specifically avoided using these parts of the textbook and openly criticized them. However, women being met with this kind of rhetoric in their textbooks is unacceptable and will surely push women out of STEM fields.

Despite common belief that scientists are objective and therefore not susceptible to bias, science faculty are often unintentionally biased against female students. In a study by Moss-Racusin et al. (2012), science faculty from research-intensive universities were given an application for a lab manager position that they believed was a real undergraduate student application. The faculty member then rated the application believing that their feedback would be going to the student. Each faculty member was given exactly one set of application materials, but there were two versions – one with a female name and one with a male name, everything else remaining identical. Despite the applications being identical, the female student on average was seen as less competent and less hirable, and she was offered a lower starting salary and less career mentorship than the identical male student (Moss-Racusin et al., 2012). It is also worth noting that female faculty were just as likely to prefer the male student as male faculty. This highlights that male faculty members are not the problem. Rather, the current culture of STEM that these biases exist within is the problem, and anybody can perpetrate these biases.

The faculty bias against female students is particularly worrisome because there is nothing that the student themselves can do to rectify the situation. While stereotype threat can be mitigated, spatial skills can be practiced, and representation can be sought out, there is nothing she can do about the subtle bias against women that exists within her professors. In the same study by Moss-Racusin et al. (2012), the faculty members said that they liked the female student more than the male student. So, a female student who is better liked and identically qualified is receiving no benefit because of her likability. This is also evidence in favour of the claim that

this bias is unintentional, and faculty members are not consciously biased against female students since they did find her to be more likeable. Unfortunately, this subtle bias has real impacts on female students such as missed job opportunities, scholarships, and mentorship (Moss-Racusin et al., 2012), the lack of which could contribute to women leaving STEM.

In my experience leading gender-based STEM initiatives, it has always seemed clear to me that simply having more women in the room would not fix the existing culture of STEM. When every single computer scientist, mathematician, engineer, or scientist you learn about is male, when almost all of your professors are male, when you are often one of a handful of female students in every single class, and when you are explicitly told that women do not belong there, it is easy to feel like you do not belong. I truly believe that the systemic underrepresentation of women in STEM can be changed. This paper demonstrates that at every point in a woman's life, she is pushed away from STEM. Despite the scientific community taking more action in addressing gender disparity in STEM in recent years, it has not been enough and there is more work that can still be done. Since female students are underperforming in spatial skills, which are an important predictor of earning a STEM degree in the future, there could be more emphasis placed on developing those skills early on. Interventions could be done with female teachers with math anxiety so that it is not passed on to her female students. There are evidence-based strategies that could be incorporated into classrooms and course materials to mitigate stereotype threat by including identity-safe cues. Finally, science faculty members could receive more education about recognizing subtle bias and how to be on guard against it.

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